

(3) Consider the transformation $w = z + (1+i)$ and determine the region in the w -plane corresponding to the ~~(triangle)~~ triangular region bounded by the lines $x=0$, $y=0$ and $x+y=1$ in the z -plane.

Sol: Given $w = z + 1+i$

$$\text{i.e., } u+iv = (x+iy) + (1+i)$$

$$\Rightarrow u = x+1 \text{ and } v = y+1$$

$$\text{or, } x = u-1 \text{ and } y = v-1$$

$$\text{Now, } x=0 \Rightarrow u=1; \quad y=0 \Rightarrow v=1;$$

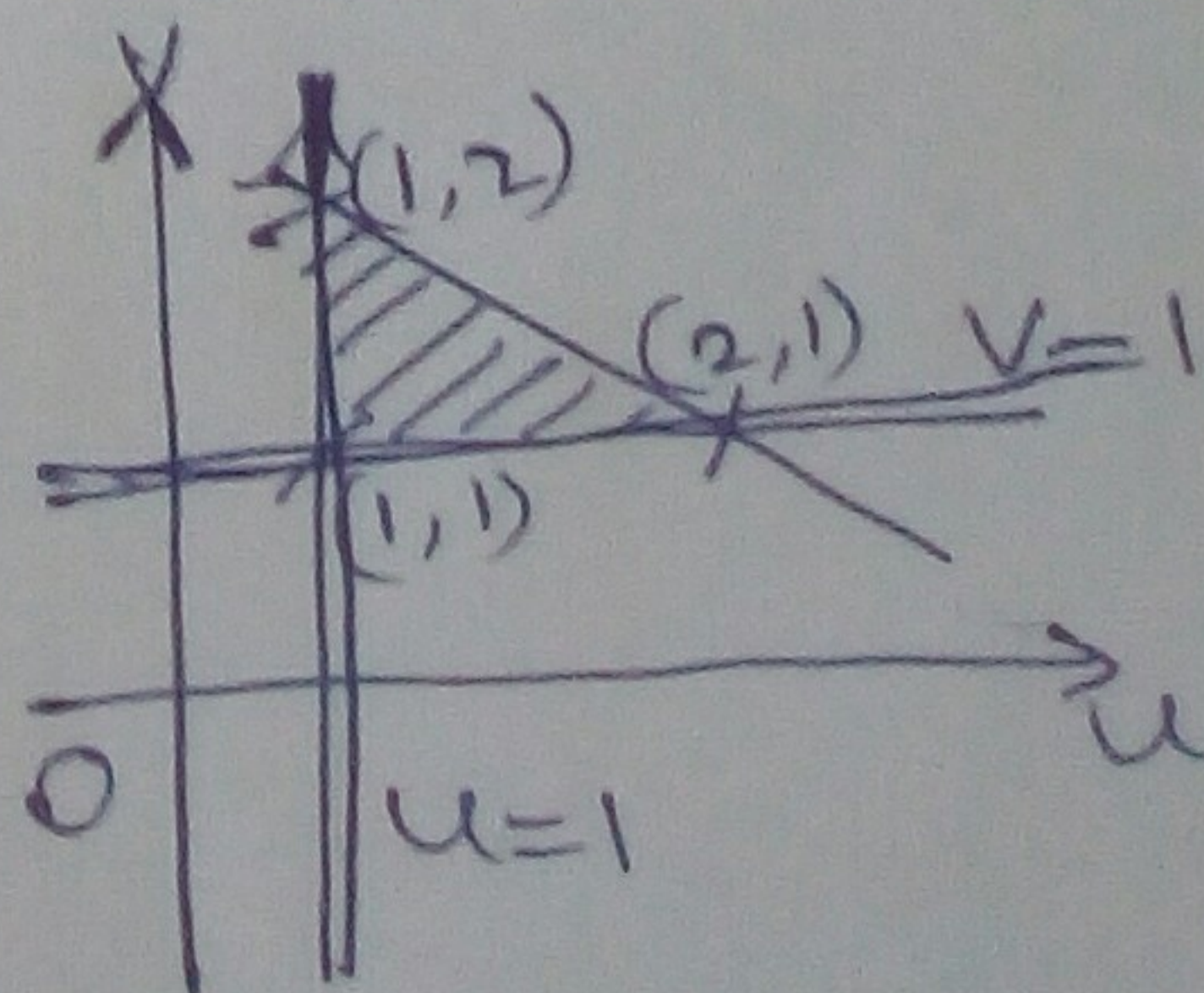
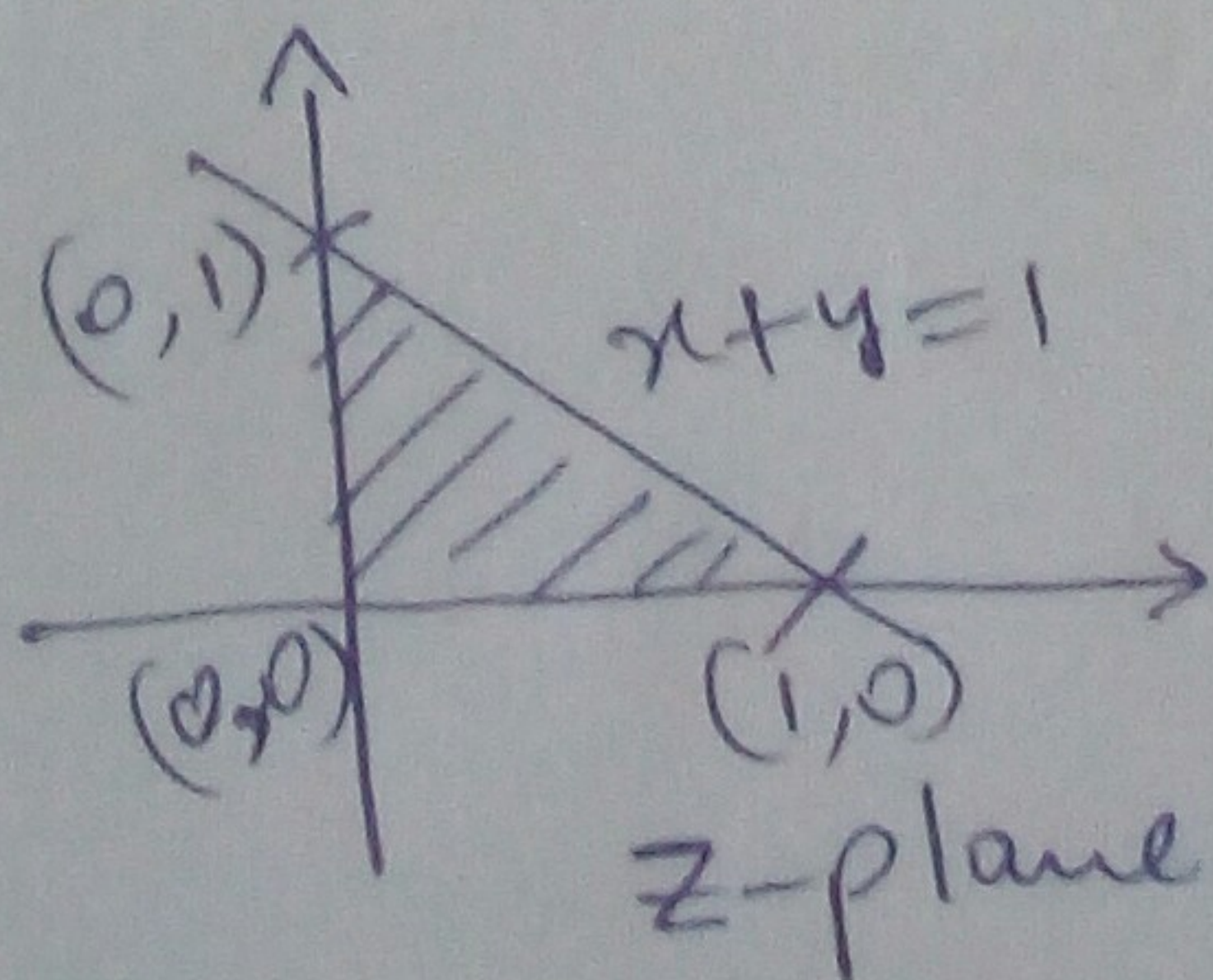
$$x+y=1 \Rightarrow (u-1) + (v-1) = 1$$

$$\Rightarrow u+v=3$$

Therefore, the lines $x=0$, $y=0$ and $x+y=1$

in the z -plane are mapped to $u=1$,

$v=1$ and $u+v=3$ respectively in w -plane.



Hence, the region becomes triangle bounded by the lines $u=1$, $v=1$ and $u+v=3$ in the w -plane.

④ Find the image of the circle $|z|=2$ under the transformation $w=z+(3+2i)$

2. Magnification and Rotation:

The transformation ~~(2.7)~~ $w = cz$, where c is a complex constant, represents both magnification and rotation.

If $w = R e^{i\phi}$, $z = r e^{i\theta}$ and $c = \rho e^{i\psi}$,

$$\text{Then } w = cz \Rightarrow R e^{i\phi} = r \rho e^{i(\theta+\psi)}$$

Here r is the magnitude of z , $r\rho$ is the magnitude of w . The amplitude of z is θ and $\theta+\psi$ is the amplitude of w .

The magnitude is multiplied by a constant and the amplitude is rotated

through an angle ψ . Therefore, this transformation is called the magnification and rotation. Note that the circle $|z|=a$

is transformed to a circle $|w|=re$.

Note: Magnification: $w = \lambda z$ ($\lambda > 0$); Rotation: $w = e^{i\alpha} z$
positive real number.

Examples:

① what is the image of the rectangular region of the z -plane bounded by the lines $x=0$, $y=0$, $x=1$ and $y=2$ under ~~the~~ the transformation $w=2z$ in the w -plane.

Sol: Given $w = 2z$

$$\text{i.e., } u + iv = 2(x + iy)$$

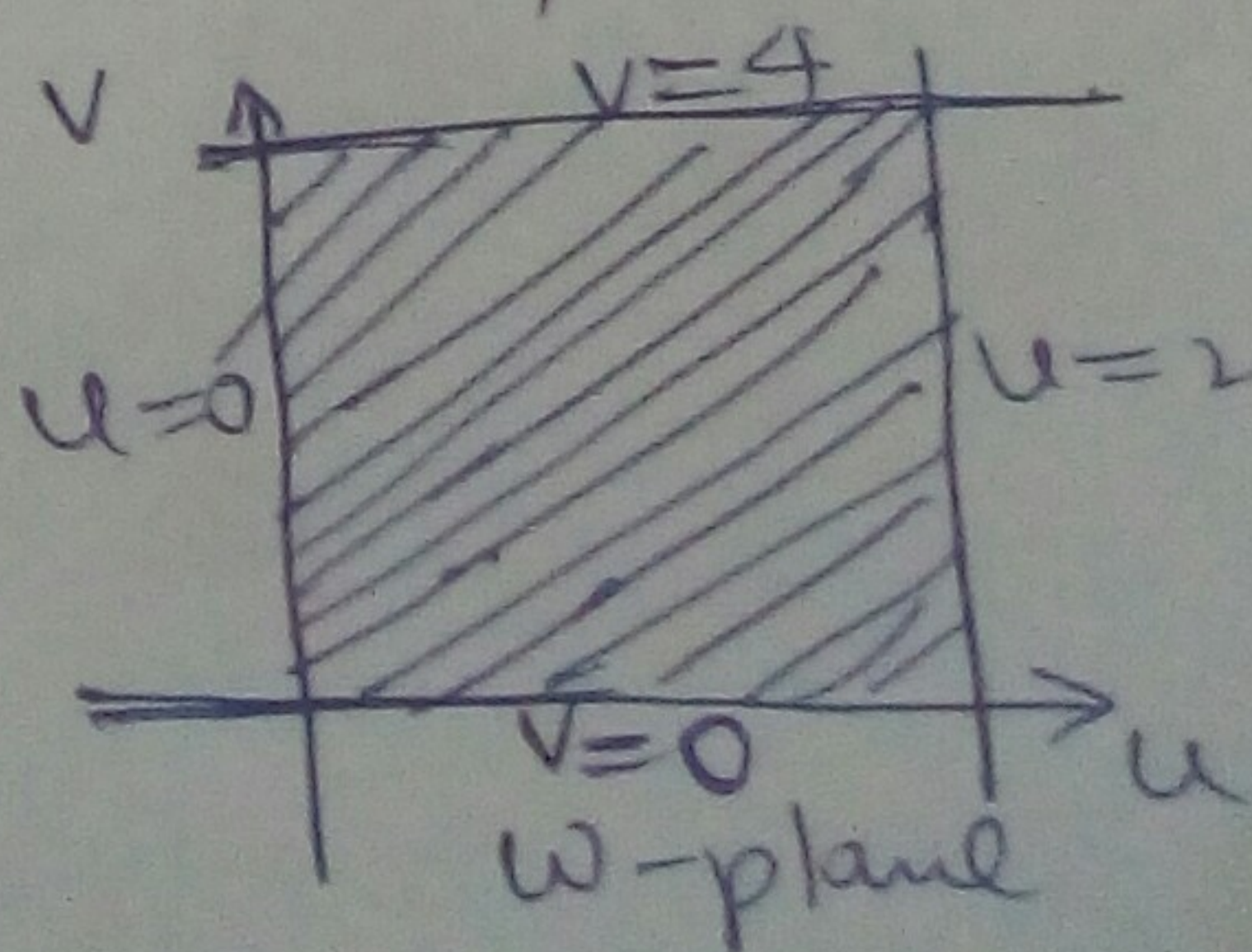
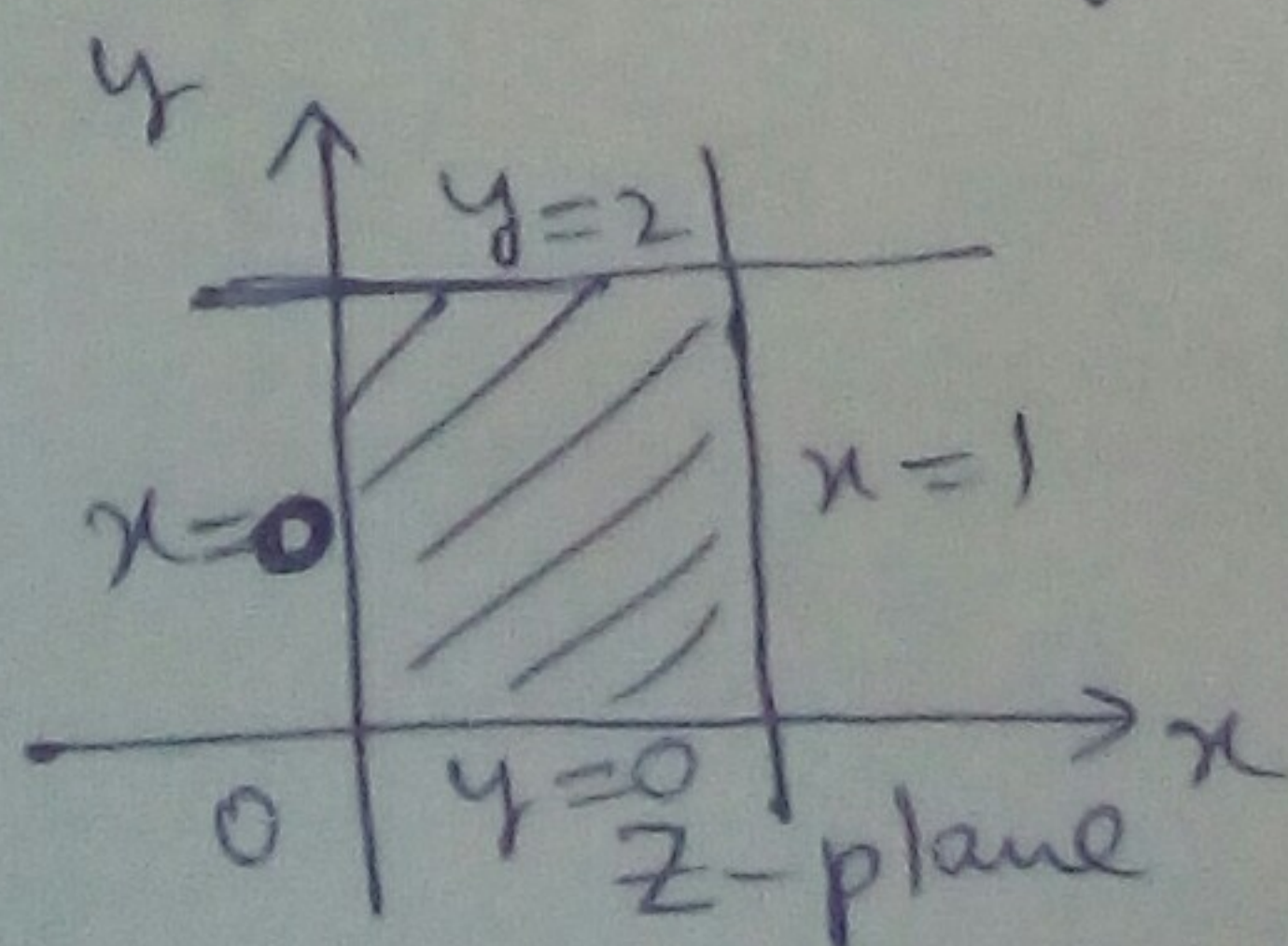
$$\Rightarrow u = 2x \text{ and } v = 2y$$

$$\text{Now, } x=0 \Rightarrow u=0$$

$$y=0 \Rightarrow v=0$$

$$x=1 \Rightarrow u=2$$

$$y=2 \Rightarrow v=4$$



Therefore, the image of the rectangle in the z -plane is the rectangle in the w -plane.

- (2) Find the image of the ~~(triangular)~~ triangular region bounded by the lines $x=0$, $y=0$ and $x+y=1$ in the z -plane under the transformation

$$w = e^{i\frac{\pi}{4}} z.$$

Sol: Given $w = e^{i\frac{\pi}{4}} z$

i.e., $u+iv = \left(\cos\frac{\pi}{4} + i\sin\frac{\pi}{4}\right)(x+iy)$

$$\Rightarrow u = \frac{x}{\sqrt{2}} - \frac{y}{\sqrt{2}} \text{ and } v = \frac{x}{\sqrt{2}} + \frac{y}{\sqrt{2}}$$

or, $\sqrt{2}u = x - y$ and $\sqrt{2}v = x + y$

Now,

$$x=0 \Rightarrow \sqrt{2}u = -y \text{ and } \sqrt{2}v = y$$

$$\text{so, } \sqrt{2}u = -\sqrt{2}v \Rightarrow v = -u$$

and $y=0 \Rightarrow \sqrt{2}u = x$ and $\sqrt{2}v = x$

$$\text{so, } u = v$$

Also, $x+y=1 \Rightarrow \sqrt{2}v = 1 \Rightarrow v = \frac{1}{\sqrt{2}}$

